ANY CONCEPCION GONZALEZ RODRIGUES

Molecular Bioengineer | Neural Electrophysiology, HD-MEA & Cell Culture

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RESEARCH PROFILE

Molecular Bioengineering graduate with research experience in neural cell cultures, co-culture models and high-density microelectrode array recordings. Conducted research at DZNE and CRTD in Dresden, including the development of an *ex vivo* retinal electrophysiology workflow, experimental optimization and neuronal activity analysis using Python and R. Interested in Organoid Intelligence, particularly how electrophysiology and feedback-driven experiments can be used to investigate learning and neural computation in human brain organoids.

EDUCATION

MSc in Molecular Bioengineering | University of Porto - ICBAS/FEUP
Sep 2022 - Oct 2024 | Porto, Portugal

Thesis: Establishing CMOS-MEA Recordings in Adult Zebrafish Retina

Supervisor: Prof. Dr. Michael Brand - CRTD, Germany

BSc in Bioengineering | University of Porto - ICBAS/FEUP
Sep 2018 - Jun 2022 | Porto, Portugal

Thesis: Development of Optical Sensors Using Molecularly Imprinted Polymers for Ferritin Detection

Supervisor: Dr. Margarida Piloto - ISEP, Portugal

RESEARCH EXPERIENCE

Research Intern – Neural Cell Culture and Co-culture
German Center for Neurodegenerative Diseases (DZNE), Dresden | Sep 2023 – Feb 2024

- Maintained olfactory cell cultures under sterile conditions and prepared olfactory bulb–olfactory epithelium co-cultures for functional studies.
- Prepared media and experimental solutions, followed culture protocols and monitored culture conditions.
- Performed MEA recordings during experimental stimulation and contributed to immunofluorescence staining and confocal microscopy.
- Documented experiments and discussed results within an international neuroscience research team.

Master's Thesis Researcher – Neural Electrophysiology
Center for Regenerative Therapies Dresden (CRTD), TU Dresden | Feb 2024 – Aug 2024

- Established and optimized an *ex vivo* HD-MEA workflow for adult zebrafish retina, combining dissection, perfusion, light stimulation and pharmacological experiments.
- Troubleshot tissue attachment, recording conditions and analysis workflows; maintained experimental documentation.
- Analysed spontaneous and stimulus-evoked activity using Python and R, including firing-rate measurements and stimulus-response analysis; presented findings in scientific figures and a master's thesis.

Research Intern
BioMark, ISEP, Porto | Feb 2022 – Jul 2022

- Contributed to optical sensor development for ferritin detection using molecularly imprinted polymers and UV–Vis spectroscopy

LABORATORY & TECHNICAL SKILLS

Cell culture and experimental biology: Aseptic culture techniques; olfactory cell cultures and co-cultures; culture maintenance; media and solution preparation; biological sample handling.

Molecular biology and imaging: DNA extraction; conventional PCR; agarose gel electrophoresis and gel imaging; immunofluorescence; confocal microscopy.

Electrophysiology and tissue preparation: MEA and HD-MEA recordings (Axion BioSystems, 3Brain); adult zebrafish retinal dissection; ex vivo tissue preparation; perfusion; visual and pharmacological stimulation; protocol optimization.

Data analysis: Python and R for neuronal activity analysis, statistical comparisons and data visualization; spike detection and sorting; firing-rate and stimulus-response analysis.

Research practices: Experimental documentation, troubleshooting, scientific figures and presentations, collaboration in international research teams.

PROFESSIONAL EXPERIENCE

Account Management Intern
MaxWell Biosystems, Zurich | May 2025 – Apr 2026

- Supported scientific communication on HD-MEA applications involving neural cultures, organoids and tissue preparations.
- Worked with scientific and commercial teams to organize technical information and support webinars and research-focused communications.

TRAINING AND CERTIFICATIONS

- FELASA-accredited laboratory animal training LAS modules C01 and F09-F11
- Zebrafish handling course
- Introduction to Python and Machine Learning for the Biosciences (i3S)

LANGUAGES

Portuguese—native (C2); English — C1; Spanish — B1; German — A1; French — A1

RESEARCH INTERESTS

- Organoid Intelligence
- Human brain organoids
- Neural plasticity and learning
- Electrophysiology and HD-MEA
- Closed-loop stimulation
- Neural computation and biological computing.

ACADEMIC REFERENCES

Prof. Dr. Michael Brand — Principal Investigator, CRTD, TU Dresden, Germany
Email: michael.brand@tu-dresden.de | Master's Thesis PI

Paul Röppenack — Research Supervisor, CRTD, TU Dresden, Germany
Email: paul.roeppenack@tu-dresden.de | Master's Thesis Supervisor